MIL-B-16924C
4 April 1972
SUPERSEDING
MIL-B-16924B
17 July 1967

MILITARY SPECIFICATION

BLOOD RECIPIENT SET, INDIRECT TRANSFUSION

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a blood recipient set suitable for the indirect transfusion of whole blood and the administration of blood fractions from standard blood collection-dispensing bags and blood bottles.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

GG-I-526	Instruments, Dental and Surgical; General Specifications.
GG-N-196	Needles, Hypodermic: for Luer Syringes.
MMM-A-250	Adhesive, Water-Resistant, (for Sealing Fiberboard Boxes).
PPP-B-585	Boxes, Wood, Wirebound.
PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	Box, Fiberboard.
PPP-T-76	Tape, Pressure-Sensitive Adhesive Paper, (for Carton Sealing).

FSC 6515

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T09:37Z
Check the source to verify that this is the current version before use.

STANDARDS

Federal

FED-STD-601 Rubber, Sampling and Testing; Method 3021.

Military

MIL-STD-105 Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129 Marking for Shipment and Storage.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions, should be obtained from the procuring activity or as directed by the contracting officer).

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

U.S. PHARMACOPEIAL CONVENTION, INCORPORATED

Pharmacopeia of the United States (U.S.P.)

(Application for copies should be addressed to the Mack Publishing Company, Easton, Pa. 18042.)

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI Z70.1 Dimension of Glass and Metal Luer Tapers for Medical Applications.

(Application for copies should be addressed to the American National Standards Institute (ANSI), 1430 Broadway, New York, New York 10018).

NATIONAL INSTITUTES OF HEALTH

Public Health Service Regulations - Part 73* Pertaining to Federal Control of the Sale, Barter, or Exchange of Biological Products.

*(Application for copies should be addressed to the Laboratory of Biologics Control, National Institutes of Health, Bethesda, Maryland 20014.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D2240-68 Standard Method of Test For Indentation Hardness of Rubber and Plastics By Means of a Durometer.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T09:37Z
Check the source to verify that this is the current version before use.

(Application for copies should be addressed to the American Society For Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19103).

U. S. DEPARTMENT OF HEALTH, EDUCATION AND WELFARE

Federal Food, Drug and Cosmetic Act and Regulations Promulgated Thereunder.

(Applications for copies should be addressed to the Food and Drug Administration, U. S. Department of Health, Education and Welfare, Washington, D. C. 20204).

3. REQUIREMENTS

3.1 Material.

3.1.1 Rubber and plastic components. All rubber and plastic components shall be of the best quality, and suitable in all respects for the intended purpose. All rubber and plastic components shall be sterilizable, free from pyrogens, compatible, and safe (nontoxic), when tested as specified in section 4. Those rubber and plastic components coming in contact with solutions being administered, shall contain not more than 5 parts per million (p.p.m.) total of lead, barium, and cadmium when tested as specified in 4.4.1. The plastics comprising the fluid path shall meet the requirements of the Pharmacopeia of the United States (U.S.P.) for Class II plastics when tested as specified in 4.4.2.

* 3.1.1.1 Molded rubber connector. The molded rubber connector shall be fabricated from latex.

3.1.1.2 Plastic tubing. Tubing shall be clean and free of particles. Plastic tubing shall be nontoxic, colorless, transparent, flexible polyvinyl, and shall meet the requirements for Class II plastics as stated in the U.S.P. The tubing shall be flexible and pliable, but sufficiently rigid to prevent collapse during normal use. The tubing shall be clean and free of particles and shall be free from all liquid lubricating substances which oxidize, exude, or sweat-out to the surface of the tubing.

* 3.1.1.3 Piercing device. The piercing device shall be fabricated from a modified polymethylmethacrylate rigid plastic.

3.1.1.4 Other plastic parts. Other plastic parts shall be fabricated of materials suitable for the intended purpose. The material shall be sterilizable and shall be nontoxic (safe) when tested as specified in 4.4.3.

3.1.2 Needles.

3.1.2.1 Cannulae of needles. Cannulae of needles shall be fabricated from 304 AISI stainless steel, and shall meet the corrosion-resistance, elasticity, and resistance-to-breakage tests as specified in Specification GG-N-196. The cannulae shall be fabricated from either seamless drawn, or welded and drawn tubing.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T09:37Z
Check the source to verify that this is the current version before use.

3.1.2.2 Hubs. Hubs shall be class 3, type 302, 303, or 304; or class 8, alloy number 260, 270, or 360 of specification GG-I-526; or hubs shall be plastic. Class 8 (brass) hubs shall be nickel plated.

3.1.3 Other material. All other materials as specified herein shall be of the best quality and suitable in all respects for the purpose intended.

3.2 Style and design. Style and design shall be in accordance with Figure 1 and as hereinafter specified.

* 3.2.1 Description. The blood recipient set shall be a disposable set consisting of a filter drip housing with drip device and filter, a piercing device (a coupler with a piercing tip), a length of plastic tubing, a molded rubber connector, a male needle adapter, an intravenous needle, end protectors, a control clamp, and an instruction sheet. The set shall provide a sterile, pyrogen-free fluid path completely sealed to maintain sterility, and shall be suitable for the indirect transfusion of whole blood and administration of blood fractions from standard blood collecting-dispensing bags (FSNs 6515-926-4766 and 6515-926-4767) and blood bottles.

3.2.1.1 Piercing device. A sharpened plastic piercing device (coupler with piercing tip) for the purpose of puncturing the blood container and firmly connecting a fluid path from the blood container to the recipient set, shall be firmly attached in a tight, leakproof manner to the upper end of the filter drip housing. The piercing device shall be a cylindrical tube, molded in one piece, 2-1/2 inches (plus or minus 1/8 inch) overall in length. The exposed portion of the piercing device shall be not less than 1-1/4 inches in overall length. The piercing device shall have a flat circular flange 3/4 inch (plus or minus 1/4 inch) in diameter, and a thickness of not less than 1/16 inch, permanently positioned on the long axis of the coupler not less than 1-1/4 inch from the piercing tip. The flange shall be so constructed as to serve as the point to apply pressure to insert the piercing tip into the standard blood bag outlet ports and blood bottle stopper, and shall be of sufficient strength for this purpose. The opposite (proximal) end of the coupler shall be inserted into the upper chamber of the housing for a distance of 5/8 inch (plus or minus 1/16 inch), and into the filter for a distance of 3/8 inch (plus or minus 1/16 inch). Both the housing and the filter shall be permanently secured to the coupler in such a manner that the coupler shall not loosen, permit leaking, or damage the housing when the piercing device is inserted into, or removed from, the blood bottle stopper, or inserted by twisting into the outlet port(s) of the standard blood collecting-dispensing bag(s) and donor set. The shaft of the piercing device shall consist of a cylindrical tube. The exposed shaft of the piercing device, measured from the distal end of the flange shoulder to the heel of the piercing tip aperture, shall be 3/4 inch (plus 3/64 or minus 9/64 inch) in length.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T09:37Z
Check the source to verify that this is the current version before use.

The outside diameter shall taper from 0.220 inch (plus or minus 0.003 inch) at the distal end of the flange shoulder to 0.203 inch (plus or minus 0.003 inch) at the heel of the aperture. The internal diameter shall taper from 0.168 inch (plus or minus 0.003 inch) at the nonexposed extremity of the piercing device (that is, end in filter) to 0.152 inch (plus or minus 0.003 inch) at the heel of the aperture of the piercing tip. The piercing tip of the coupler shall be in the form of a sharp half cone. A single, regular-shaped orifice on the side opposing the half cone shall furnish the pathway for the fluid. The effective area of this orifice shall be not less than 0.018 square inch and shall permit passage of a 0.126 inch (minimum) diameter ball throughout the lumen of the piercing device. The piercing tip shall be sharpened to such an edge as to be suitable for the puncture of, and insertion into the port of a standard plastic blood bag, without damaging the bag, or into the blood bottle stopper shown in Figure 2, without exertion of undue pressure. The piercing device or the tip of the piercing device shall not break or fracture when placed perpendicular to the 1 centimeter thick portion of a standard rubber stopper when gentle but firm pressure is applied. The construction of the piercing device shall make a snug, leak-proof fit with the blood bottle stopper and the outlet ports of the plastic blood bag for the duration of the use of the recipient set, and shall permit the recipient set to be removed from the stopper or ports of the bag without difficulty and without damage to the stopper or the bag (that is, except for the puncture). The piercing device shall be ruggedly designed and structurally sound. It shall not break, or leak air or fluid during use. The piercing device shall not be brittle or soft and shall not become brittle or soft under normal storage conditions.

3.2.1.1.1 Durability of piercing device. The durability of the piercing device shall be tested as specified in 4.4.4. The piercing device shall be so firmly attached to the filter housing that the juncture of the device and the housing shall not loosen or permit leaking, nor shall the device or housing become damaged. The piercing device shall be capable of being inserted into the blood containers (into the blood bottle rubber stoppers, Figure 2, and the outlet ports of the blood bag), as described in 4.4.4, without the device breaking or bending to the degree of causing difficulty when properly inserted into the blood container.

* 3.2.1.2 Filter-drip housing. The filter drip housing shall be fabricated from transparent, colorless, flexible, semi-rigid, polyvinylchloride plastic having a hardness of 98 plus 2, minus 4 Shore A at 25° C. when tested as specified in 4.4.6. The housing shall be flexible enough to permit squeezing by hand without undue effort, yet rigid enough to withstand the collapse test as specified in 4.4.5.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T09:37Z
Check the source to verify that this is the current version before use.

The filter drip housing, on the finished unit, shall be sturdy enough to withstand normal manipulation. After insertion of piercing device into port of bag the flexibility of the drip chamber shall allow it to be 1/4 filled with blood by a slight squeeze and release. After insertion of piercing device into the stopper of blood bottle the flexibility of the filter chamber shall allow it to be completely filled with blood by a slight squeeze and release. The filter drip housing shall consist of two separate chambers so arranged that the upper chamber contains the filter and the lower chamber, the outlet of the drip device, and designed in such a manner that the fluid must first pass through the filter before flowing through the drip device, when the set is in its operating position. The filter shall be permanently secured to the top of the filter chamber and the coupler (piercing device), in such a manner that the fluid shall flow outward through the filter into the upper chamber then through the drip device of the lower chamber when the set is in operation. The fluid flow through the filter must progress from the inside to the outside of the filter. The filter shall be approximately cylindrical in shape, closed at its bottom end, and supported directly below the connector so that fibrin or clots, passing through the connector into the filter, tend to settle to the bottom without flowing down over the walls of the filter. The filter material shall be of nylon mesh. The mesh shall be a 70 to 100 mesh sieve (sieve number as defined in the U.S.P. for Standard Sieves), and shall have a filtering area of 5.25 inches plus or minus .8 square inch. The size of the filter chamber shall be such as to allow the use of the filter specified above, and shall not restrict the filter so as to impede the flow of fluid through the filter. The drip chamber shall contain a drip device, the free length of which shall be not less than 5 millimeters (mm). The size of the drip chamber shall be such as to accommodate a volume of not less than 8 milliliters (ml.), and there shall be a minimum distance of 1-3/8 inches from the end of the drip device to the bottom of the drip chamber. The drip device shall be so constructed and calibrated that the fluid flowing into the lower chamber shall be broken into drops, 15 such drops, plus or minus two drops, shall equal 1 ml. (Twenty drops of water, at 15° Centigrade (°C), shall weigh 1.25 grams (gm), plus or minus 0.1 gm). The bottom of the drip chamber shall have an orifice through which the plastic tubing shall be permanently attached, by heat sealing, to the chamber housing.

* 3.2.1.3 Plastic tubing. Plastic tubing shall be no less than 60 inches in length, measuring 0.116 inch (plus 0.015 inch, minus 0.010 inch) in inside diameter and 0.020 inch (plus or minus 0.005 inch) in wall thickness. One end of this tubing shall be permanently and securely heat sealed, to form an air-tight, leakproof connection with the orifice in the lower chamber of the housing. The other end of the tubing, internally reinforced with a 1/2 inch (plus or minus 1/8 inch) length of rigid plastic tubing, shall be inserted

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T09:37Z
Check the source to verify that this is the current version before use.

into a molded latex rubber part at the connector proximal to the shouldered center section of the molded part at the connector proximal to the shouldered center section of the molded part. This shall be accomplished in such a manner as to form an air-tight, leakproof connection which shall not separate or leak when tested as specified in 4.4.7. The inside diameter of the reinforcing tube shall be no less than twice the inside diameter of the needle.

* 3.2.1.4 Molded rubber connector. The molded latex rubber connector shall be 1-1/2 inch, plus 3/16 inch, minus 1/16 inch in length, 0.050 inch, plus or minus 0.015 inch in wall thickness at the connector ends and shall be constructed with a raised center section, diameter of which is nominal .500 inch, length 5/8 inch plus or minus 1/16 inch and a minimum wall thickness of .060 inch. The center section shall be shouldered and thickened at one end to provide for easy insertion of an 18 gage needle cannula when adding required supplemental medication, and the shoulder shall be of such thickness as to be capable of resealing subsequent to puncture by a nonsiliconized 18 gage needle. Shall withstand an internal fluid pressure of at least 7 p.s.i. This test shall be accomplished by capping the end of tubing and raising the internal pressure to 7 p.s.i. This molded part shall facilitate checking veni-puncture by the flashback technique. The molded rubber part shall have inserted in the distal connector a rigid plastic needle adapter which shall form an air-tight, leakproof connection (see 3.2.1.5). When tested for resealing as specified in 4.4.8, there shall be no falling drop within 10 seconds.

* 3.2.1.5 Needle adapter. The needle adapter shall be a rigid, suitable plastic cylindrical tube approximately 1-1/2 inches in length, and so constructed as to be attached to the molded rubber part in a leakproof manner (see 3.2.1.4). The other end of the adapter shall be tapered to fit a needle having a standard Luer hub in accordance with ANSI Standard Z70.1 and shall form an airtight leakproof connection (see 3.5). The adapter, or assembly of the adapter to the molded part, shall be such as to prevent accidental reversal of the assembly of the adapter to the rubber component at time of assembly. The needle adapter shall be tested for dimensions as specified in 4.4.9. Needle adapters with Luer tapered ends which do not fit the test gage within specified limits shall be considered not acceptable. In addition, when the test gage is fitted onto the Luer tapered end, as specified in 4.4.9, there shall be no movement on the tapered end when the gage is pressed lightly from side to side in all directions. The needle adapter shall make an air-tight, leakproof connection with the molded rubber part which shall not separate or leak. When tested as specified in 4.4.7, there shall be no falling drop within 10 seconds.

* 3.2.1.6 Intravenous needle. The needle shall be a 18 gage, 1-1/2 inch, regular bevel, intravenous needle. Except as specified herein, the needle shall be in accordance with the requirements of Specification GG-N-196 for type I, point style a, hub style 1 or 2 needle, except that marking may be omitted. The cannula and hub shall be fabricated of material as specified in 3.1.2.1 and 3.1.2.2, respectively. The hub cavity shall be constructed in accordance with figure 3 of ANSI Standard Z70.1. The needle shall fit the needle adapter specified in 3.2.1.5 in such a manner as to

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T09:37Z
Check the source to verify that this is the current version before use.

make a tight, leakproof connection. The hub cavity shall be tested for diameter and depth and fluid tightness as specified in 4.4.10. Hubs shall fit the plug gage within the "GO" - "NO GO" limits of the feeler gage as shown on figures 5 and 6 of ANSI Standard Z70.1. Hubs which do not fit the plug gage within these limits shall be considered not acceptable. In addition, when the hub is fitted onto the plug gage, as specified in 4.4.10.1, there shall be no movement of the hub on the gage when the cannula is pressed lightly from side to side in all directions. When tested for fluid tightness, as specified in 4.4.10.2, the swaged connection of the cannula with the hub, and the connection of the hub with the test gage, shall not show leakage exceeding the formation of a falling drop. The needle shall be supplied with hub attached to the tapered end of the needle adapter or in approximation for such attachment and shall not be assembled to the set by placing the point in the inverted position in the distal end of the needle adapter. The needle and needle adapter shall be covered by a transparent plastic protector in such a manner as to maintain the sterility and integrity of the entire fluid pathway of the infusion device until the protector is manually removed.

* 3.2.1.7 End protectors. End protectors shall be readily removable transparent plastic tubes or sheaths. The end protectors shall be placed over each open end of the infusion device, and shall maintain sterility, freedom from pyrogens, immobility of the needle, and integrity of the entire fluid path of the infusion device, until manually removed.

3.2.1.8 Control clamp. The control clamp shall be either the screw type or the roller type, and shall be fabricated from suitable plastic. The control clamp shall be attached in open position on the plastic tubing and shall be moveable on the tubing. The clamp shall provide a constant setting and quick shutoff from the operating position, and shall be capable of one-hand operation. The design shall be such that the screw or roller cannot be accidentally removed. To assure constant delivery of fluid, it shall be impossible for the initial clamp setting to be changed by a sudden hard pull on the tubing. The control clamp shall not cut or damage the tubing in either the open or closed position. When tested as specified in 4.4.11, the clamp shall be capable of accomplishing complete quick shutoff (that is, no falling drop in drip chamber or from needle adapter), shall regulate flow of the fluid from complete shutoff to maximum flow, and with a constant head of flow. In addition, when the clamp is operated from its fully open position to complete shutoff, and this operation is repeated ten times while the clamp is in the same position on the tubing, the tubing shall not be cut, pierced, or damaged so as to cause leakage or restriction of flow of fluid.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T09:37Z
Check the source to verify that this is the current version before use.

* 3.2.1.9 Assembly. The needle adapter (3.2.1.5) shall be connected to one end of the molded rubber part (3.2.1.4). The other end of the molded rubber part shall be connected to the plastic tubing (3.2.1.3), and the control clamp (3.2.1.8) set on the tubing. The filter drip housing (3.2.1.2) shall be connected to the plastic tubing. A piercing device with base flange (3.2.1.1) shall be attached or integrally molded to the filter drip housing. The needle (3.2.1.6) and cannula shall be assembled to the set. The end protectors (3.2.1.7) shall then be fitted to the set. The end protectors shall contain a cotton vent to maintain sterility or they shall be sealed after sterilization.

* 3.2.1.10 Instruction sheet. Each set shall be provided with a detailed instruction sheet containing ample illustrations, and clear, concise instructions for proper use of the set with the standard plastic blood bags and a stoppered vacuum blood bottle. Instructions shall include a conversion table permitting an estimation of the number of milliliters infused per minute based on observed number of drops falling from the drip device in a 15-second interval. The table shall also include a method for computing the total quantity of blood infused in a given time (and vice versa).

3.3 Compatibility. The materials of which the sets are manufactured shall not react with, or be acted upon, by the liquids with which they will be in contact during normal use and operation, nor shall there be any materials incorporated in the set which will cause any harmful or deleterious effect when the set is used for the purpose intended.

* 3.4 Sterility, toxicity, and freedom from pyrogens. All equipment shall be nontoxic (safe). Fluid paths, end connectors, and needles of sets shall be sterile and pyrogen-free. All the above shall be in accordance with the additional standards: Whole Blood (Human) of the Public Health Service Regulations - Part 73. The fluid path, end connectors and needles shall be pyrogen-free, when tested as specified in 4.4.12, and sterile when tested as specified in 4.4.13. The sets shall be nontoxic (safe) when tested for safety as specified in 4.4.3.

3.4.1 Certificates. For destination inspection only, certificates of sterility, freedom from pyrogens, and safety, as specified below, shall be submitted by the contractor directly to Technical Operations Division, Directorate of Medical Materiel, Defense Personnel Support Center, 2800 South 20th Street, Philadelphia, Pa. 19101. Attention: Quality Assurance Branch, and one copy of each shall be forwarded with each shipment to the consignee.

3.4.1.1 Certificate of sterility. A certificate of sterility, listing all sterility lot or control numbers, shall state that samples from each sterility lot or control have been tested for sterility, and were found to be sterile. The certificate shall reference the method used in determining sterility.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T09:37Z
Check the source to verify that this is the current version before use.

3.4.1.2 Certificate of safety. A certificate of safety, listing all lot or control numbers, shall state that samples from each lot or control have been tested for safety and were found to comply with such requirements. The certificate shall reference the methods used in determining safety.

3.4.1.3 Certificate of freedom from pyrogens. A certificate of freedom from pyrogens, listing all lot or control numbers, shall state that samples from each lot or control have been tested for pyrogens, and were found to comply with such requirements. The certificate shall reference the method used in determining freedom from pyrogens.

* 3.5 Leakage of set. Sets shall be tested for leakage as specified in 4.4.7. Sets which show leaking, loosening, or separation of any part of the set shall be rejected. A leak is defined as the formation of an air (gas) bubble which breaks away from the set within the specified test time (5 seconds).

3.6 Separation of set. The piercing device or the recipient tube shall not separate, or loosen, from the filter-drip housing when tested as specified in 4.4.14.

3.7 Filter and drip chamber collapse Sets shall be tested for filter and drip chamber collapse as specified in 4.4.5. Sets in which filter or drip chamber collapse sufficiently to empty fluid from drip chamber and allow air to enter recipient tube shall be rejected.

3.8 Clear passage. Sets shall be tested for clear passage as specified in 4.4.15. Sets which are sealed or constricted shall be rejected.

* 3.9 Manufacturer's record. The manufacturer shall place a code or lot number on the paper label, or on the printed carton, which can be verified against the manufacturer's records of all tests made on nontoxic, sterile, pyrogen-free sets. These records shall be made available for inspection by any duly qualified medical officer of the Armed Forces, National Institutes of Health, or any official of the Federal Food and Drug Administration, or any duly authorized Government representative. No shipment shall be made until all the tests have been completed, verified, and recorded in the permanent records of the manufacturer.

* 3.10 Government furnished material. The Government shall furnish the contractor the plastic blood bags (FSN 6515-926-4767) needed to perform the piercing device durability test specified in 4.4.4, and the clamping device test specified in 4.4.11.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T09:37Z
Check the source to verify that this is the current version before use.

3.11 Workmanship. Workmanship shall be first class throughout. All materials, treatments, and procedures used in the fabrication and assembly of the recipient sets shall be such that the finished sets shall not show any evidence of decomposition, deterioration, or other conditions which may adversely affect the appearance or serviceability of the sets at time of delivery or during storage.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

* 4.1.3 Certificates of quality. Certificates of quality supplied by the manufacturer of the rubber, plastic and steel may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the FSN, the item identification, the name of the component/material, the lot number, the lot size, the sample size, the date of testing, the test method, the individual test results, and the specification requirements.

4.2 Sampling.

* 4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105. The inspection levels and acceptable quality levels shall be as indicated in Table I. The unit of product for examination shall be one blood recipient set.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T09:37Z
Check the source to verify that this is the current version before use.

Immediate container shall be marked to identify contents:

Table I. Sampling for Examination

Type of defect	Inspection level	AQL (percent defective)
Major	II	1.0
Minor	II	2.5

* 4.2.2 For tests.

* 4.2.2.1 For sterility test. Sampling for the sterility test shall be in accordance with the United States Pharmacopeia (USP).

4.2.2.2 For pyrogen, and safety tests. Sampling procedures for pyrogen, and safety tests shall be conducted as indicated in the U.S.P., for transfusion and infusion assemblies.

4.2.2.3 For heavy metals test. Samples shall be taken from each lot of rubber or plastic as indicated in paragraph 4.4.1.

* 4.2.3 For other tests. Sampling for other tests shall be conducted in accordance with MIL-STD-105, and as indicated in Table II.

Table II. Sampling for Other Tests

Component (unit of product)	Characteristic	Reqmt.	Test procedure	Insp. level	AQL (percent defective)
Rubber and plastic components	Heavy metals (extractable)	3.1.1	4.4.1	**	*
1-Set	Biological	3.1.1	4.4.2	**	*
1-Set	Safety (toxicity)	3.1.1.4	4.4.3	***	*
1-Set	Piercing device durability	3.2.1.1.1	4.4.4	S-3	*
1-Set	Filter & drip chamber collapse	3.2.1.2 & 3.7	4.4.5	S-3	*
1-Set	Housing hardness	3.2.1.2	4.4.6.1	S-3	*
1-Set	Filter area	3.2.12	4.4.6.2	S-3	*
1-Set	Leakage of set	3.5	4.4.7	I, except sample size shall be a minimum of 125 samples	*
1-Set	Reseal	3.2.1.4	4.4.8	S-3	*
1-Set	Needle adapter	3.2.1.5	4.4.9	S-3	*
1-Set	Intravenous needle	3.2.1.6	4.4.10	S-3	*
1-Set	Clamping device	3.2.1.8	4.4.11	S-3	*
1-Set	Freedom from pyrogen	3.4	4.4.12	***	*
1-Set	Sterility	3.4	4.4.13	***	*
1-Set	Separation of set	3.6	4.4.14	S-3	*
1-Set	Clear passage	3.8	4.4.15	S-3	*

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T09:37Z
Check the source to verify that this is the current version before use.

* Acceptance number shall be zero for all sample sizes. Minimum of two samples shall be taken regardless of lot size.

** Sufficient sample units to perform the tests shall be selected from each lot but each lot of product need not be tested provided evidence is furnished that the formulation had previously been tested, and that it is the same as being approved on this contract.

*** Sampling for tests shall be on component lots.

* 4.2.4 Sample unit usage. Where possible, the same sample unit shall be used for the determination of two or more test characteristics.

4.2.5 For filter drip housing hardness test. Sufficient samples shall be selected from each lot to perform the test. Failure to comply with test requirements shall be cause for rejection of the lot represented.

* 4.3 Examination. The sets shall be examined to determine compliance with all requirements contained in this specification. Examinations shall be conducted in accordance with MIL-STD-105 and the classification of defects specified in Table III. Inspection, however, shall not be restricted to the classified possible defects listed in Table III. Any sets in the sample containing one or more defects shall be rejected and if the number of defective sets in any sample exceeds the acceptance number for the sample, the lot represented by the sample shall be rejected.

Table III. Classification of Defects

Categories	Defects
Critical	None defined.
Major	
101	End protectors missing or dislodged.
102	Closed type protector not free of holes, or cotton missing (or inadequate amount) from open type protector.
103	Tubing and filter housing not free of bloom or tackiness.
104	Set incomplete.
105	Style and design not as specified.
Minor	
201	Packaging, packing and marking not as specified.
202	Instructions missing, incomplete or illegible.
203	Contractor's name or registered trademark missing, incomplete, or illegible.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T09:37Z
Check the source to verify that this is the current version before use.

4.4 Tests.

* 4.4.1 Heavy metals-extractables test. A 2 gram composite sample shall be obtained from the rubber and plastic components. The material shall be cut into small slivers, so that no thickness exceeds 3 mm., and eluted for 72 hours at 50° C., 24 hours at 70° C., or 60 minutes at 121° C. in two menstrua: 0.9% sodium chloride and cottonseed oil. Each gram of the material shall be covered with 5 milliliters (ml.) of the menstruum. The menstrua are collected, agitated after elution, and then submitted to spectographic analysis for lead, barium, and cadmium. The metals are reported in parts per million. Thus, 1.0 ppm in elute is equivalent to extracting 1.0 mg. from 200 g test material dispersed in 1,000 ml. of elution menstruum.

* 4.4.1.1 Alternate heavy metals-extractable test. Accurately weigh a 5 gram sample for each of the metals (barium, cadmium or lead) into a 1 oz. Vycor crucible. Reduce to ash in a muffle furnace at 650° C. for eight hours. Let cool to room temperature and add 15 ml of 1:1 hydrochloric acid. Bring to a boil on a hot plate, allow to boil for 2 to 3 minutes, remove, let cool, and filter through Whatman #2 filter paper into a 50 ml volumetric flask. Rinse the residue on the filter with deionized water and dilute to 50 ml. Prepare standard solutions of 1, 2, and 5 ppm for each of the metals, namely barium, cadmium, and lead. Determine the absorbances of these solutions at appropriate wave lengths using a Hollow Cathode Bu, Cd, or Pb lamp for each of the respective metals and a 10 x scale expansion on a spectrophotometer, such as the Perkin-Elmer Model 301 Atomic Absorption Spectrophotometer. Determine the absorbance of the sample solutions using the same conditions as with the standards.

Calculations:

$$\frac{A \times F \times \text{dilution}}{W} = \text{ppm barium, cadmium, or lead.}$$

A = Absorbance

$$F = \text{Factor} = \frac{(\text{concentration standard in ppm Ba, Cd, or Pb})^*}{\text{Absorbance}}$$

W = Sample weight in grams.

* The three standards are averaged to give ppm/unit A.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T09:37Z
Check the source to verify that this is the current version before use.

4.4.2 Biological tests. The plastics shall be tested in accordance with the U.S.P. for Class II plastics.

4.4.3 Safety (toxicity) test. Samples of the finished sets shall be tested for safety, in accordance with U.S.P., for transfusion and infusion assemblies.

4.4.4 Piercing device durability test. The piercing device (on assembled set) shall be inserted into, and removed from, a blood bottle as shown on Figure 2 and inserted and twisted into, and removed from, a plastic blood bag (FSN 6515-926-4767), in the normal manner. This operation shall be repeated twice. While the piercing device or tip of the piercing device is being inserted into the stopper perpendicular to the 1 centimeter thick portion of the standard rubber stopper, a gentle but firm pressure shall be applied to the set. The piercing device shall then be immersed in Anti-coagulant Solution "A" in accordance with the U.S.P. and the Additional Standards: Whole Blood (Human) of the Public Health Service Regulations - Part 73, for 4 hours. The piercing device shall then be removed from the solution, and immediately inserted in a plastic blood bag as specified above.

4.4.5 Filter and drip chamber collapse test. The filter chamber shall be filled to one-quarter its capacity, the drip chamber to one-third its capacity, and the recipient tube shall be completely filled with water at 100° F.. Close off the entrance to the piercing connector lumen. Hold set in a vertical position, with needle adapter three feet below the bottom of the drip chamber. Remove end protector from needle adapter and allow fluid to flow from open end of set. Subject to this condition for one minute.

* 4.4.6 Housing test.

4.4.6.1 Housing hardness. From each blend lot, mill a slab of plastic sheeting. From this milled slab take a sample of approximately 25 gm and press this sample into a panel 3.0 inches by 5.0 inches by 1/8 inch. Allow panel to stand for eight hours and then determine hardness using a durometer. Each blend lot shall meet the hardness specified in 3.2.1.2. As an alternate, shall be in accordance with ASTM D2240-68.

4.4.6.2 Filtering area. The length of filter below the heat seal shall be measured and the filter area calculated. The filtering area shall be as specified in 3.2.1.2.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T09:37Z
Check the source to verify that this is the current version before use.

* 4.4.7 Leakage of set test. Samples of recipient sets (sterilized) shall be subjected to a compressive force between the flange of the piercing device and the filter chamber (i.e., normal to flange of piercing device) of 25 pounds for 2 seconds. A torque of 4 inch pounds shall be exerted, for 2 seconds, between the piercing device and the filter chamber. A separation force of 8 pounds for 2 seconds shall be exerted between the piercing device and recipient tube. The set shall then be immersed in water and subjected to an internal pressure (pneumatic pressure) of 8 pounds per square inch gage (p.s.i.g.) for 5 seconds. This test shall be performed with a gage as shown in figure 8 of ANSI Standard Z70.1 (and having a suitable fluid path throughout the length of the gage) fitted to the Luer tapered tip of the needle adapter.

* 4.4.8 Reseal test (molded rubber part). This test shall be applied to those samples previously subjected to the leakage test (4.4.7). The assembled set shall be connected to a full, suitably vented, one liter bottle of water, saline solution or dextrose solution. The bottle shall be hung in a normal manner so that the liquid level in the bottle shall be three feet above the injection site, and the set shall be adjusted to a drip rate of 25 drops (plus or minus 2 drops) in 15 seconds. The test shall be completed before the bottle is one half empty. While the solution is flowing, the molded rubber part shall be placed in a horizontal position and shall be punctured from the underside in the shouldered section with an 18 gage non-siliconized needle and the needle immediately withdrawn. After puncture, the tubing shall be clamped, or otherwise closed off, and the molded rubber part shall be subjected to an internal pressure (hydrostatic pressure) 7.0 pounds per square inch gage (p.s.i.g.) for 30 seconds and shall meet the requirement specified (see 3.2.1.4).

4.4.9 Needle adapter. The tapered end of the needle adapter shall be tested for proper dimensions and taper by means of the ring gage shown in figure 7 of ANSI Standard Z70.1. This gage shall be slipped onto the tip with moderate pressure.

4.4.10 Intravenous needle.

4.4.10.1 Diameter and depth of hub cavity. The needle hub cavity shall be tested on the male, Luer taper gage in accordance with plug gage of figure 5 and 6 of ANSI Standard Z70.1. (This gage shall be as shown in ANSI Standard Z70.1, except that it shall be constructed with a suitable fluid path to permit performance of the fluid tightness test specified in 4.4.10.2). The hub shall be slipped onto this gage with very light pressure. The fit of the hub on this gage shall be tested by means of the feeler gage in accordance with figures 5 and 6 of ANSI Standard Z70.1.

4.4.10.2 Fluid tightness. With the needle in a horizontal position, the hub shall be attached by means of a plug gage, as shown in figure 5 of ANSI Standard Z70.1 (except that gage shall have a fluid path), to a water pressure source having a pressure gage. The water shall be run through the needle so as to fill any hub cavity space and the lumen of the cannula, the lumen then sealed by a piece of solid rubber slipped onto the point, and a pressure of 45 pounds per square inch applied for 10 seconds.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T09:37Z
Check the source to verify that this is the current version before use.

4.4.11 Clamping device test. The assembled set (except for needle) shall be connected, in the normal manner, to a plastic blood bag (see 3.10) which has been filled with water and sealed against leaking. The blood bag, with set connected thereto, shall be suspended in the normal manner, at a sufficient height to permit the entire length of the tubing of the set to hang vertically. The ability of the clamp to accomplish complete shut off, to regulate flow, and, with a constant head of water, to maintain a constant rate of flow, shall be tested with the clamp at a point on the plastic tubing within two inches above the molded rubber part. The tests shall be performed with the tubing hanging vertically at all times. The rate of flow shall be determined either by counting the drops in the drip chamber during a measured time period, or by recording the time required to collect a measured volume of fluid from the needle adapter.

4.4.12 Freedom from pyrogen. Samples of the finished sets shall be tested for freedom from pyrogens, in accordance with the U.S.P., for transfusion and infusion assemblies.

* 4.4.13 Sterility. Samples of the finished recipient sets shall be tested for sterility, in accordance with the procedures for parenteral devices as described in the U.S.P.

4.4.14 Separation of set test. This test shall be applied to those samples previously subjected to the leakage test (4.4.7). Exert a straight pull separation force between the piercing device and the filter drip housing, of 30 pounds for 2 seconds. Exert a separation force between the filter-drip housing and the recipient tube of 10 pounds for 2 seconds.

4.4.15 Clear passage test. Samples, after sterilization, shall be subjected to an internal water pressure of 3 pounds per square inch.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2).

5.1.1 Unit of issue. One package, containing forty-eight (48) sets, as specified, constitutes one unit of issue.

5.1.2 Packaging quantities. The number of units of issue indicated in Table IV shall be packaged in each unit, intermediate, and exterior container, as applicable, for the required level of protection specified in the procurement document.

Table IV. Packaging Quantities

Unit package	Intermediate package	Exterior container
1 unit	Not Required	2 units

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T09:37Z
Check the source to verify that this is the current version before use.

5.1.2.1 Packing variation permitted. If the required number of units in the entire shipment is less than the number of units specified to be overpacked in an exterior container, such units may be packed in an exterior container of suitable size and design, acceptable to a common carrier, which shall insure safe delivery to destination.

5.1.3 Level A.

5.1.3.1 Individual box. Each set shall be packaged in a sealed box, except commercial colors will be acceptable. Box shall be furnished with perforated thumb notches, located at each end of the box, to facilitate opening. The complete unit shall be packaged in such a manner as to prevent the tubing from being kinked or taking a permanent set.

5.1.3.2 Unit package. Unit package shall be a box of appropriate size and design constructed in accordance with PPP-B-636, type CF, class weather-resistant. Boxes shall be closed by taping the top and bottom flaps with 2 inch wide tape conforming to PPP-T-76. Closing of bottom flap with adhesive conforming to MMM-A-250 is permitted. Metal stitches are not permitted.

5.1.4 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Level A.

* 5.2.1.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class weather-resistant. Closure and strapping shall be as specified in the appendix of the applicable box specification. Special requirements as specified in the applicable table of PPP-B-636 shall apply for fiberboard boxes.

5.2.2 Level B.

* 5.2.2.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, class domestic. Closure shall be as specified in the appendix of the applicable box specification. Fiberboard box shall be closed by method II of PPP-B-636. Special requirements as specified in the applicable table of PPP-B-636 shall apply for fiberboard boxes.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T09:37Z
Check the source to verify that this is the current version before use.

5.3 Marking.

5.3.1 Individual box. Each box shall be marked in accordance with the requirements of the Federal Food, Drug and Cosmetic Act. Each box shall contain instructions as specified in 3.2.1.10. Marking shall include the legend: "FLUID PATH AND END CONNECTORS ARE STERILE AND PYROGEN FREE" or similar statement.

5.3.2 Unit package. Each unit package shall be marked in accordance with MIL-STD-129. When labels are utilized, waterproofing shall be required only when applicable box is fabricated of water-resistant material. Lot (control) number, contract or purchase order number and name of contractor shall be shown. In addition, marking shall include the method of sterilization (G-Gas, R-Radiation, S-Steam, D-Dry Heat Sterilization) shall be shown following the level of packaging and date of packaging. Example: "A-3/72-G". Final sterilization shall be on or after indicated packaging date.

* 5.3.3 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129. Lot (control) number shall be shown. The method of sterilization as specified in 5.3.2 shall also be shown.

NOTE: Each lot number shall be traceable to a specific sterilized charge.

6. NOTES

6.1 Intended use. The blood recipient set is intended for indirect transfusion of whole blood and administration of blood fractions from standard blood collecting-dispensing bags and donor sets and blood bottles.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
(c) Federal stock number.

* 6.3 This specification is intended to cover the following item appearing in the Federal Supply Catalog: FSN 6515-089-3470, BLOOD RECIPIENT SET, INDIRECT TRANSFUSION, Disposable w/needle, 48s. It does not include all types, sizes, or grades as indicated by the title nor all types which are commercially available.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T09:37Z
Check the source to verify that this is the current version before use.

* 6.4 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and Contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

Custodians:
Army - MD
Navy - MS
Air Force - 03

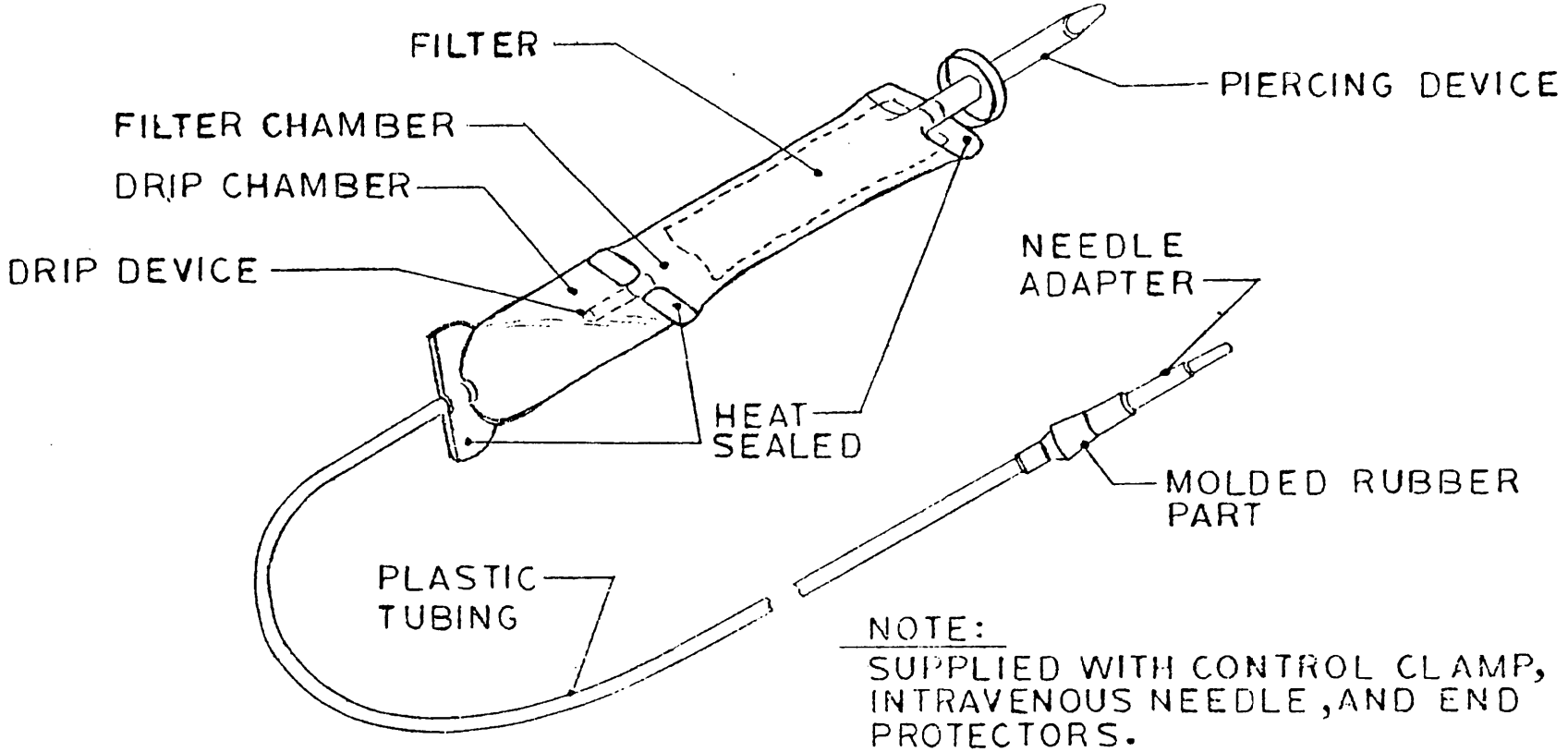
Review activities:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Project No. 6515-1481

Review information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current DODISS.

Source: https://assist.dla.mil -- Downloaded: 202610-18T09:37Z
Check the source to verify that this is the current version before use.



NOTE:
SUPPLIED WITH CONTROL CLAMP, INTRAVENOUS NEEDLE, AND END PROTECTORS.

FOR DETAILED REQUIREMENTS SEE APPLICABLE PARAGRAPHS OF SPECIFICATION

STYLE DESIGN AND ASSEMBLY OF SET PROPER

FIG. 1
BLOOD RECIPIENT SET

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T09:37Z
Check the source to verify that this is the current version before use.

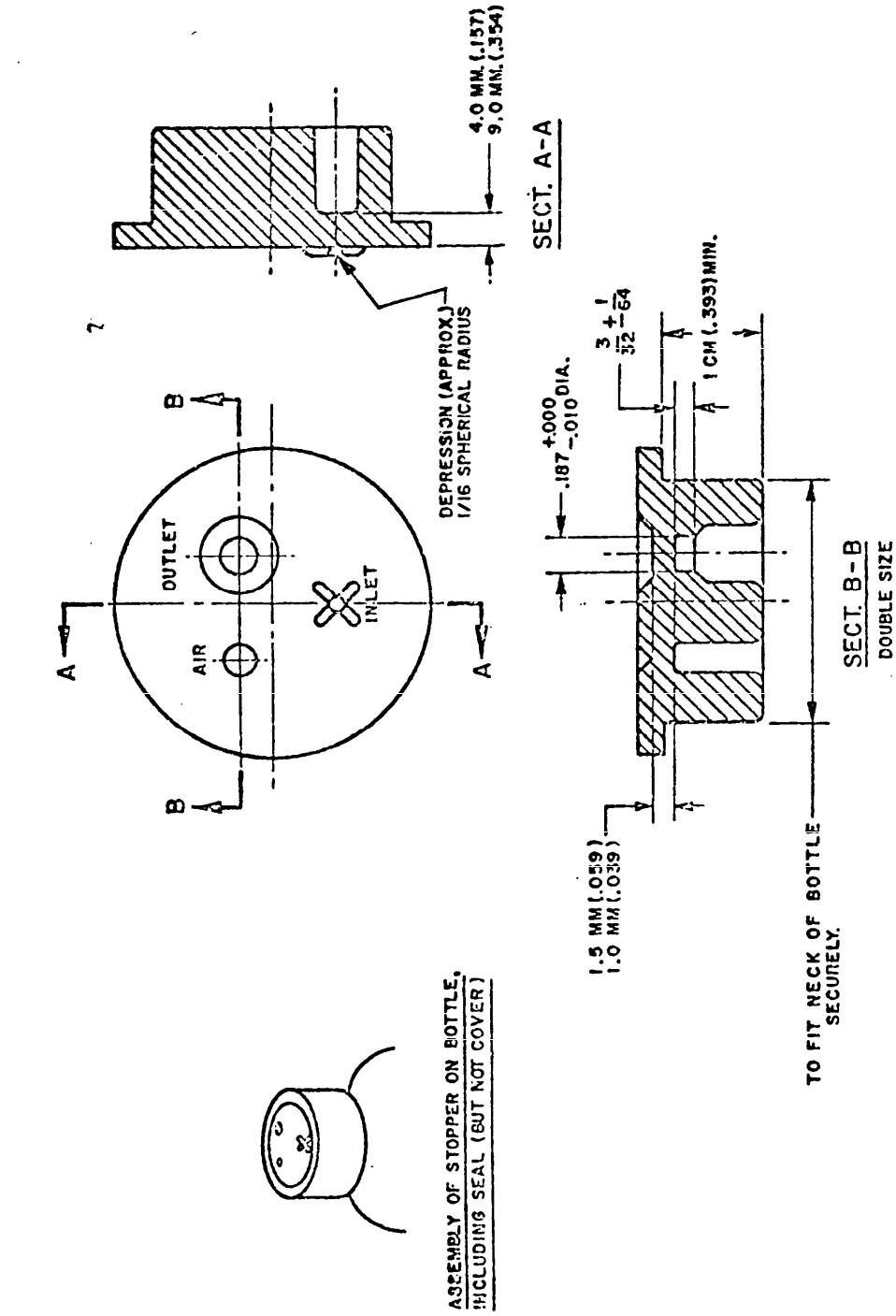


FIG. 2

Source: https://assist.dla.mil -- Downloaded: 202610-18T09:37Z
Check the source to verify that this is the currentversion before use.

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION
MIL-B-16924C

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64, WHICH MAY BE USED.

Source: https://assist.dla.mil -- Downloaded: 202610-18T09:37Z
Check the source to verify that this is the current version before use.

FOLD

DEFENSE PERSONNEL SUPPORT CENTER
2800 South 20th Street
Philadelphia, Pa. 19101

Defense Supply Agency

POSTAGE AND FEES PAID
DEFENSE SUPPLY AGENCY

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

HEADQUARTERS, DEFENSE PERSONNEL SUPPORT CENTER
ATTN: DIRECTORATE OF MEDICAL MATERIEL, CODE ATT
2800 SOUTH 20TH STREET
PHILADELPHIA, PA. 19101

FOLD

Source: https://assist.dla.mil -- Downloaded: 202610-18T09:37Z
Check the source to verify that this is the current version before use.